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Events and Phases(rev.3)

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Introduction

A trajectory simulation is a prime example of a "time-event" simulation. The key concepts needed to build a successful time-event simulation are those of events and phases, and their application in the simulation world. This memo focuses on rocket trajectory simulations, but these concepts have a much broader region of applicability.

Events

Events are important milestones in a trajectory. There are two kinds of events, organic and adaptive. Organic events are characteristic of the rocket itself and how it is operated. They are often known a priori as functions of time after liftoff (TALO). A pair of organic events defines the boundaries of most trajectory phases. Organic event examples are solid propellant rocket burnout (in the absence of closed-loop thrust termination) and parachute deployment. On the other hand, adaptive events arise from interactions among the rocket, its environment and its trajectory. For these, the event times are not known a priori. Good examples are apogee and impact. While adaptive events are often not phase boundaries, transition from constrained motion on a launcher rail to free flight is an adaptive event dependent on distance traveled.

Event Monitoring

All the events encountered in a rocket trajectory simulation depend on a single variable. For example, apogee depends on the vertical component of velocity and second stage ignition depends on TALO. A standard technique for detecting these is the "sign test" applied to the product of the key variable in two adjacent time steps. To detect apogee one checks the value of the product of vertical velocity in two adjacent time steps. In all cases but one this product will be positive. When the product is negative, we know that the apogee event lies somewhere inside the relevant time step. Sometimes a bias value must be used in the product variables. For example to detect impact at some altitude h_I one forms the product of $(h - h_I)$ from two adjacent time steps and seek that pair giving a negative product.

Given that we know the times bounding an event, it's usually possible to construct a simple linear interpolator to refine the estimate of the interesting variable. Consider apogee as an example. During the brief integration time step, it's a good approximation to consider the vertical acceleration to be constant. Then, the vertical velocity V_z within the key time step is

$$V_z = V_{z1} - g(t - t_1).$$

Here the subscript $()_1$ denotes the value of $()$ at the start of the key time step. From this, the time of apogee $t^*(V_z = 0)$ can be estimated as

$$t^* = t_1 + V_{z1}/g.$$

Apogee altitude h^* can then be estimated from

$$h^* = h_1 + V_{z1}(t^* - t_1) - g(t^* - t_1)^2/2 = h_1 + \frac{V_{z1}^2}{2g}.$$

Similar results can be readily found for other events. One trick will be needed when one interpolates to find the value of γ at an adaptive event. There will be no estimate of $d\gamma/dt$ developed in the code because γ should not be a state vector element. Estimate it from the intrinsic coordinate result,

$$d\gamma/dt = -g\cos\gamma/V_{TOT}.$$

Here
γ = Flight path angle, positive when the velocity vector points above the horizon,
g = Acceleration due to gravity, and
V_{TOT} = Total (vector magnitude) velocity.

Another kind of adaptive event are maxima and minima for variables not elements of the state vector. For such variables their derivatives are not readily available. Good examples are maximum dynamic pressure and maximum Mach number. For these events the variable x_i is monitored and when $x_{i+1} < x_i$ the maximum has been approximately found. A first estimate on the maximum is, of course, x_i.

If a refined estimate is needed, fit a parabola to x_{i-1}, x_i & x_{i+1}. The dependent variables are separated by the time step, Δt. Taking the time to be zero at x_{i-1},

$$x = x_{i-1} + At + Bt^2,$$

where $A = -\frac{x_{i+1} - 4x_i + 3x_{i-1}}{2\Delta t}$, and $B = \frac{x_{i+1} - 2x_i + x_{i-1}}{2\Delta t^2}$.

Differentiate and set the result equal to zero to the find the time T of the optimum:

$$T = -\frac{2A}{B}$$

Given the optimum time, it is plugged into the parabolic formula to obtain the desired stationary value of the dependent variable.

Phases

Phases are what happens between phase boundary events. Simulation phases are described as logical variables. A simulation run requires that each phase in the Table 1. menu must be input as either TRUE or FALSE by the user. Once a set of phases has been selected TRUE, the simulation can execute the TRUE phases in the order in which they appear in the menu.

Most importantly it should be possible to mix and match phases from this list to accomplish one's simulation objectives. That is, any given simulation run need not use all these phases. If the rocket does not have a parachute recovery system, for example, then the last two phases will not be needed (they should be selected FALSE). Second, this list should be considered incomplete. It does not, for example, include free flight, both powered and coasting, with a spin-stabilized stage because this technique has never been used in a small sounding rocket[1].

Since all phases except first stage powered gravity turn are initiated on TALO, a simple approach is to include in the input table the Sequence of Events with initial TALO for each phase, and, if different from the end state of the previous phase, the initial conditions on mass, altitude, velocity and flight path angle/quadrant elevation angle (QE).

Every phase is unique, characterized by

- Its initial criterion. This is usually TALO or some such. Note that some initial times are user specified as delay times from the end of a prior event. Also, the transition from constrained motion on a launcher rail to unconstrained free flight motion is based on distance travelled.
- Initial weight (mass). Although mass is often included as an element in the state vector (a good idea), its initial value at the start of a new phase will depend on the mass dropped before starting the new phase.
- Dynamic model. There are three of these: Constrained motion on the launcher rail, Free flight, and Free flight with a temporary decrease in integration time needed for initial parachute deployment and inflation.
- Thrust and drag models. Every powered flight phase is associated with a thrust-time table. See Thrust Curve.org. All coasting flight phases are associated with a power-off drag curve and all powered flight phases are associated to a power-on drag curve. Aerodynamic reference areas are included as part of this description. Often, parachute phases use the uninflated parachute area as the reference area because that's the way the parachute literature is usually written.

[1] Ordinary spin stabilized ordnance, e.g., a rifle bullet, can be adequately modeled with the first stage coasting gravity turn after burnout phase. However, for sounding rocket flight at altitudes too high for practical aerodynamic stabilization spin stabilization is usually required. The dynamic model for this includes some tricky precession effects.

It is important to note that if an event does not change any of the above characterizations, it is not a phase boundary event. Apogee and maximum dynamic pressure, both important events, are not phase boundary events. On the other hand, burnout is usually a phase boundary event because it is often associated with staging.

For a simulation capable of addressing a three stage rocket the phases to be included are shown in Table 1 below:

Description	I	II	III	IV	V
First Stage on Coasting Flight	T	T	F	F	F
First Stage Coasting on Parachute	T	F	F	F	F
First Stage Powered Flight on Launcher	F	F	T	T	T
First Stage Powered Free Flight	T	F	T	T	T
First Stage Coasting Flight	F	F	T	F	F
Second Stage Coasting Flight	F	F	F	T	T
Second Stage Powered Free Flight	F	F	F	T	T
Second Stage Coasting Flight	F	F	F	T	F
Third Stage Coasting Flight	F	F	F	F	T
Third Stage Powered Free Flight	F	F	F	F	T
Third Stage Coasting Flight	F	F	F	F	T
Coasting on Drogue Parachute	T	F	T	T	F
Coasting on Main Parachute	T	F	T	T	F

Table 1 Simulation Phase Menu

Several comments on this menu are in order. The Description row refers to a set of 5 examples:

Example I describes the sequence of events for a single stage rocket that has been dropped from an aircraft, parachute-stabilized before ignition, and then recovered with two parachutes. After ignition the rocket flies up through the parachute, shredding it in the process.

Example 2. is the sequence for a bomb dropped from an aircraft.

The third Example is for another single stage rocket launched from a rail on the ground and then recovered with a two stage parachute system. Note that a single stage parachute recovery system initially deployed in a reefed condition, and then disreefed at a later time can be simulated with the same two phases with disreefing occurring when the "Main Parachute Deployment" event happens.

Example 4 is a rail-launched two stage rocket in which the simulation follows the second stage, not the expended first stage, after coasting. If we wished to follow the expended first stage, we'd use Example 3. It has a recovery system.

The fifth Example is a rail-launched three stage rocket following the third stage which has no recovery system.

The most straightforward way to implement all this in SKYAERO, written in EXCEL, is to put TALO in the first column. The next thirteen columns are for the thirteen phases listed above. The logical contents of each cell are set to "TRUE" if the simulation is in that phase at the indicated TALO. Otherwise, the default value remains "FALSE". A simple logical test on the times in the Sequence of Events will establish the correct logical value for each of the 13 cells at every TALO. Note that it's a good idea to start and end all phases concurrently with the beginning/end of an integration time step. Keep it simple. These logical variables can also be used to establish phase initial conditions if they are not the same as the ending conditions from the previous phase. Finally, given that the phase logical variable for every TALO at the start of an integration step is known, other logical variables can be used to ensure the appropriate drag curve, aerodynamic reference area, etc. are used.

Ending Events

Events ending a phase also come in two kinds, natural and timed. Examples of natural ending events are burnout (thrust = 0) and impact (altitude = h_I). The relevant phase can not continue for natural reasons. Timed events, on the other hand, end when the simulation user wants them to end. In every case, except the very first event, a new phase starts at the same time as its predecessor ends.

Thrusting events always end when vacuum thrust decays to zero. The user can easily determine the duration of a thrusting phase by inspection of its thrust-time table. The only exception is thrusting while on a launcher rail. Coasting and parachute events are all defined by user-specified durations, except for the final event ending with ground impact. For example, second stage ignition ends a second stage coasting flight phase, but only when the user wants it to happen. Thus, the user inputs the duration of all active phases except for the two ending with natural events, thrusting on the launcher rail and the final phase ending with ground impact.

When executing a trajectory code such as SKYAERO, timed phase boundaries are input from Table 2 below:

Phase Description	Duration, sec	Start TALO, sec	End TALO, sec	Logic Error?

Table 2 Duration Input Table

The first column is the Phase Description as provided in the Table Menu. . The second is the input Phase Duration. The third & fourth columns are derived by addition from column 2. The last column is a an error message. For example, if a negative duration is inadvertently input. Table 1 contains a row for each phase selected TRUE in Table 2. For the two events ending in a natural event, the duration cell is left blank.

Note that the final phase, which will be terminated by impact with the ground, should be given an input duration of 9999.0 seconds. The actual impact will occur much earlier, but this ensures that the simulation run does not stop itself prior to impact.

IMPORTANT! It's sometimes possible to input the same quantity twice. Unless care is taken, it's only a matter of time until the two numbers don't match. For example, the time of first stage burnout / start of the first stage coasting flight appears in the Sequence of Events table used, and sets the phase logical variables. It is also the last entry in the first stage table of thrust vs. time. The best approach is to fill in the Sequence of Events table using the thrust vs. time data.